INCH-POUND

MIL-H-52297D
28 November 1988
SUPERSEDING
MIL-H-52297C
9 January 1981

MILITARY SPECIFICATION

HOLDFAST ASSEMBLY: BAR TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a holdfast assembly and its components to be used with a trestle assembly (MIL-T-52150).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

AMSC N/A
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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; Appropriate Test Methods.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government drawings. The following other Government drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

Drawings

ME

TA13207E9737	- Holdfast Assembly: Bar Type.

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE- , Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN WELDING SOCIETY

AWS A5.1	- Carbon Steel Covered Arc-Welding Electrodes.
AWS A5.18	- Carbon Steel Filler Metals for Gas Shielded Arc Welding.
AWS D1.1	- Structural Welding Code, Steel.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJune Road, P. O. B ox 351040, Miami, FL 33126).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualification.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The holdfast assembly and components shall be as shown on top assembly TA13207E9737, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the (nomenclature). Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vender to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.3.1 Material deterioration prevention and control. The holdfast assemblies shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the holdfast assemblies may be exposed.

3.3.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the holdfast assemblies may be newly fabricated from recovered materials to the maximum extent practicable, provided the holdfast assemblies produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the holdfast assemblies.

3.4 Welding electrodes. Welding electrodes shall conform to AWS A5.1 or A5.18, classification and size as required.

3.5 Identification marking. The holdfast assembly or components, as applicable shall be metal die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.6 Treatment and painting. The portions of the (item) normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.7 Stenciling. The nomenclature, Federal stock number, weight and cube shall be stenciled on each holdfast assembly. The letters "NET WT" shall precede the numerals in the weight and letters "CU" shall precede the cubic displacement. The words "2 SOLDIER CARRY" shall be stencilled on each holdfast assembly. Marking shall be Gothic-type capitals and Arabic numerals of a legible size but not less than 1/2-inch high. Stenciling shall conform to MIL-T-704.

3.8 Workmanship.

3.8.1 Steel. Steel used in fabrication shall be free from rust and corrosion. Flame cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. Precaution shall be taken to avoid overheating. Maintain proper cooling rates to preclude adverse effect to the properties of the metal. Burned surfaces of flame-cut material shall be ground or machined to remove heat-effected material. These precautions are necessary to ensure end use applications. External surfaces shall be free from burrs, sharp edges, and corners, except when sharp edges or corners are required of where they are not detrimental to safety.

3.8.2 Welders and welding.

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3.8.2.1 Welders. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by one of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 Structural Welding Code - Steel
ASME Boiler and Pressure Vessel Code, Section IX.

Welding procedures used for qualification tests shall be in accordance with the contractor's welding procedures. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator. The test results shall be made available for review by the contracting officer or the contracting officer's representative.

3.8.2.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shield metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the American Welding Society Publication "D1.1 - Structural Welding Code." All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and adjoining bass metal shall be wire brushed clean. Weld procedures shall be developed in accordance with AWS D1.1 and weld procedures shall be made available to the contracting officer or his representative upon request. Each welding procedure to be used to a welding station shall be available at that station for reference.

3.8.3 Holes. All holes shall be drilled to full size or drilled under-size and reamed to full size through steel templates. All burrs and projections shall be removed from holes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies

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submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article holdfast assembly or components, as applicable, shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AWL shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The holdfast assembly or components, as applicable, shall be examined for the following defects:

Major

101. Material not as specified.
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes, or treatment.
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the holdfast assemblies.
106. Components missing or not as specified.
107. Workmanship not as specified.
108. Welder/Welding operator qualification not as specified.
109. Welding not as specified.
110. Any part or component not in accordance with the QAP requirements as shown on the drawings.

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Minor

201. Identification marking incorrect, missing, or illegible.
202. Treatment and painting not as specified.
203. Stenciling not as specified.
204. Dimensions other than those identified by QAP data not as specified.

4.6 Inspections of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose in inspection, a completely processed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

111. Containers not as specified for level A or B. Each incorrect container shall constitute one defect.
112. Blocking and bracing to prevent movement not provided as specified for level A or B.
113. Marking missing, illegible, incorrect, or incomplete.

5. PACKING

5.1 Packing. Packing shall be level A, level B, or level C as specified.

5.1.1 Level A. Each complete holdfast assembly shall be packed in a close-fitting box conforming to PPP-B-621, class 2, style 2, or PPP-B-601, overseas type, style optional. Blocking and bracing to prevent movement and damage shall be provided and shall be in accordance with MIL-STD-1186. Box closure and strapping shall be as specified in the applicable box specification, or the appendix thereto, except the strapping shall be flat and the finish shall be "B".

5.1.1.1 Holdfast pins. When procured separately from the holdfast assembly, the pins, coated with corrosion-preventive compound as specified on the drawing, shall be packed together in close-fitting boxes conforming to PPP-B-621, class 2, style 2, or PPP-B-601, overseas type, style optional. Blocking and bracing to prevent movement shall be provided in accordance with MIL-STD-1186. Box closure and strapping shall be specified in the applicable box specification, or the appendix thereto, except the strapping shall be flat and the finish shall be "B".

5.1.2 Level B. Each complete holdfast assembly shall be packed as specified for level A in 5.1.1, except the box shall be domestic type or class as applicable. Strapping is not required to be finish B.

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5.1.2.1 Holdfast pins. Holdfast pins procured separately from the holdfast assembly shall be packed as specified for level A, except the boxes shall be domestic type or class as applicable. Strapping is not required to be finish B.

5.1.3 Level C. Each complete holdfast assembly shall be packed in a close-fitting wood container in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings. The container shall comply with the applicable carrier rules and regulations.

5.1.3.1 Holdfast pins. When procured separately from the holdfast assembly, the pins, coated as specified in the drawing, shall be packed together in a close-fitting wood container in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings. The container shall comply with the applicable carrier rules and regulations.

5.2 Marking. In addition to any special or identification markings required by the contract or purchase/delivery order (see 6.2), each shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The holdfast assembly is intended for use to provide ground anchorage for the trestle assembly.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Time frame required for submission of the first article model (see 3.2).
d. Level of packing required (see 5.1).
e. Any special marking (see 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.5 Definitions.

6.5.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of this specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6 Subject term (key word) listing.

Holdfast assembly
Trestle assembly

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - ME

Preparing activity:

Army - ME

Project 5420-0164

Review activity:

Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-52297D	Holdfast Assembly: Bar Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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